INTERNAL RESEARCH · MEMO · NOT INVESTMENT ADVICE · AI-ASSISTED

Twist Bioscience

2 July 2026 · the qualitative

$102.25

NASDAQ : TWST · Mature-Company DCF

$6.37B mkt cap · 62M sh · $172M cash, zero debt · $377M revenue FY25 (+20%, GM 50.7%); FY26 guide $442-447M; adj-EBITDA breakeven guided Q4 FY26; Rallied ~+150% over 52wk to ~14x sales while still adj-EBITDA-negative; SBC-heavy (owner-FCF below the headline); ~$172M net cash; Atlas (DNA data storage) spun out May'25 → external call option

THE CENTRAL QUESTION

Twist tripled over 52 weeks to ~14x sales while still adj-EBITDA-negative — does the platform grow into the price, or has the multiple run past the fundamentals?

Twist Bioscience runs a silicon-based DNA-synthesis platform — miniaturized, high-throughput DNA writing — selling into SynBio and NGS. FY2025 revenue was $377M (+20%), gross margin 50.7%, adj-EBITDA -$46.9M and improving; management guides adj-EBITDA breakeven for Q4 FY2026 as the new Wilsonville factory ramps margins. At ~$87.5 (~$5.4B cap, net cash) the stock has run ~150% over a year to ~14x sales while still losing money, and SBC keeps owner-FCF below the adj-EBITDA headline. The DCF asks whether 16-24% growth and the margin ramp grow into that price. Weighted fair value $43.95, ~50% below spot — only the bull and ultra-bull clear the quote.

PROBABILITY-WEIGHTED EXPECTED VALUE

$43.95 expected fair value today

-57.0% vs spot $102.25

FORWARD COMPOUNDED VALUE

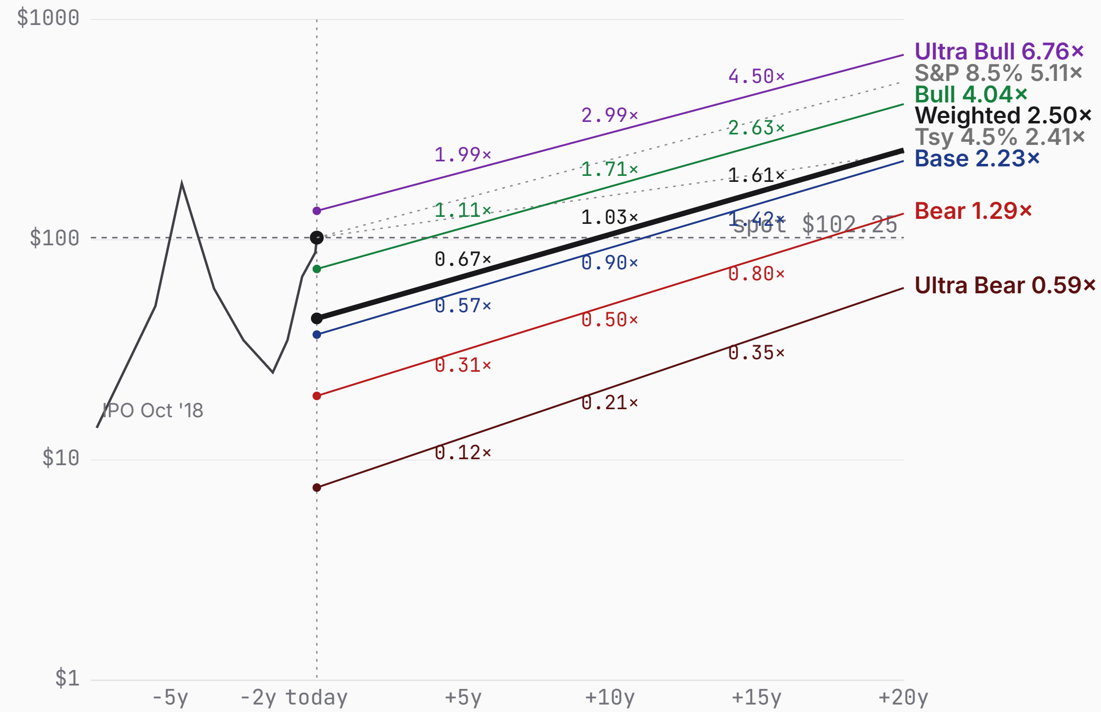
+5y	+10y	+15y	+20y
$68.15	$105.75	$164.22	$255.20
0.67× spot	1.03× spot	1.61× spot	2.50× spot

WEIGHTING RATIONALE

Ultra Bear 13% Stall at ~7.6% growth; never durably profitable.
Bear 27% Decelerate to ~12%; margins up but multiple ran ahead.
Base 33% ~16% growth + factory margin ramp; modal, ~58% below spot.
Bull 18% Hold ~20% growth; cost-leader operating leverage.
Ultra Bull 9% Dominant utility + Atlas; the only case above spot.

Bull (18%) + Ultra Bull (9%) together contribute $25.44 — 58% of the $43.95 expected value despite 27% combined probability. Today's spot ($102.25) sits 133% above the weighted expected; forward-weighted value crosses spot between +5y and +10y.

PRICE + PROJECTIONS · HISTORICAL THROUGH PROBABILITY-WEIGHTED FORWARD



ULTRA BEAR	BEAR	BASE	BULL	ULTRA BULL
$7.50	$19.55	$37.12	$73.69	$135.31
-92.7% vs spot	-80.9% vs spot	-63.7% vs spot	-27.9% vs spot	+32.3% vs spot
CAGR +7.6% WACC 11.0%	CAGR +12.2% WACC 10.0%	CAGR +16.2% WACC 9.5%	CAGR +20.1% WACC 9.0%	CAGR +24.1% WACC 8.5%
CAGR +7.6% Prob 13%	CAGR +12.2% Prob 27%	CAGR +16.2% Prob 33%	CAGR +20.1% Prob 18%	CAGR +24.1% Prob 9%
Growth stalls; never profitable	Decelerates; multiple ran ahead	Solid growth; factory margin ramp	High growth; cost leadership	Dominant utility; Atlas windfall

AR 2EB | ALAMEDA RESEARCH 2: ELECTRIC BOOGALOO
LONG HORIZONS · STRUCTURAL SHIFTS · IMAGINATION

Twist Bioscience scenario narratives

PROBABILITY WEIGHTS

Ultra Bear 13% Bear 27% Base 33% Bull 18% Ultra Bull 9% · Weighted expected $43.95 (-57.0% vs spot)

ULTRA BEAR $7.50

-92.7% vs spot

Probability 13%

Growth stalls; never profitable

WHY 13%

13% — assumes commoditization plus a funding winter that stalls the synthesis franchise. Possible given well-capitalized rivals, but cuts against 13 straight sequential-growth quarters.

WHAT HAPPENS

Revenue compounds only ~7.6% as price competition from deep-pocketed IDT and GenScript caps the synthesis franchise and biotech-funding softness slows orders. The factory's margin ramp arrives but is competed away, so owner-FCF starts negative and barely crosses into positive late in the window.

Owner-FCF margin grinds from about -8% to 8% — real but thin. At a ~7.6% grower that never builds durable platform economics, the DCF supports $7.50, roughly 91% below spot. The 14x-sales multiple was pricing an outcome the business does not reach.

BEAR $19.55

-80.9% vs spot

Probability 27%

Decelerates; multiple ran ahead

WHY 27%

27% — the natural deceleration path as a 20% grower matures into a low-teens grower against capable competition. The single most defensible outcome alongside the base.

WHAT HAPPENS

Growth slows to ~12.2% as the easy share gains mature and competitive pricing pressures the ramp. The Wilsonville factory still drives real gross-margin gains and the company reaches breakeven, but the second-derivative disappoints relative to what ~14x sales embedded.

Owner-FCF margin climbs from about -3% to 15%, generating genuine cash flow over time. But a low-teens grower at that terminal margin is worth $19.55 on the DCF, roughly 78% below spot — the business executes, the entry multiple does not survive contact with decelerating growth.

BASE $37.12

-63.7% vs spot

Probability 33%

Solid growth; factory margin ramp

WHY 33%

33% — the modal path: 13 sequential growth quarters, the guided breakeven, and a factory built to lift margins. Carries the most weight precisely because it is the management plan working.

WHAT HAPPENS

Revenue compounds ~16.2% — consistent with the raised FY26 guide of $442-447M extended forward — as Twist holds share across SynBio and NGS and the Factory of the Future delivers its designed margin step. Adj-EBITDA breakeven in Q4 FY2026 converts into sustained, growing operating leverage.

Owner-FCF margin ramps from roughly 0% to 21%, so the platform finally throws off real cash. Yet the modal value is $37.12, about 58% below spot. This is the crux — even the company executing its plan well leaves a wide gap, because the price already capitalized the success and then some.

BULL $73.69

-27.9% vs spot

Probability 18%

High growth; cost leadership

WHY 18%

18% — requires growth to hold at 20%+ AND the factory to convert into 29% owner-FCF margins. A demanding but coherent cost-leader outcome; the steelman the price needs.

WHAT HAPPENS

Growth holds ~20.1% as silicon synthesis cements its picks-and-shovels cost advantage and Twist takes share at the expense of higher-cost methods. NGS Applications scales and Protein Solutions adds a higher-value mix, pulling gross margin and operating leverage up together past breakeven into clear profitability.

Owner-FCF margin reaches about 29% as fixed factory costs lever across a much larger revenue base. A durable 20% grower with platform margins supports $73.69, still ~16% below spot — the bull case nearly justifies the quote but does not fully clear it.

ULTRA BULL $135.31

+32.3% vs spot

Probability 9%

Dominant utility; Atlas windfall

WHY 9%

9% — stacks utility-grade dominance, 24% growth, and an Atlas commercialization that is currently an external option Twist only holds equity in. Low-probability by construction; the right tail, not the expectation.

WHAT HAPPENS

Twist becomes the default DNA-synthesis utility — the silicon platform's cost and throughput advantage compounds into share dominance across research and applied markets, with platform-grade margins. Growth compounds ~24.1% and the Atlas Data Storage equity stake, spun out in May 2025, commercializes into a real call-option windfall.

Owner-FCF margin reaches ~35% as a dominant utility with pricing power. This combination supports $135.31, about 55% above spot — the only scenario that materially beats the price, and it requires both the utility outcome and the data-storage option paying off.

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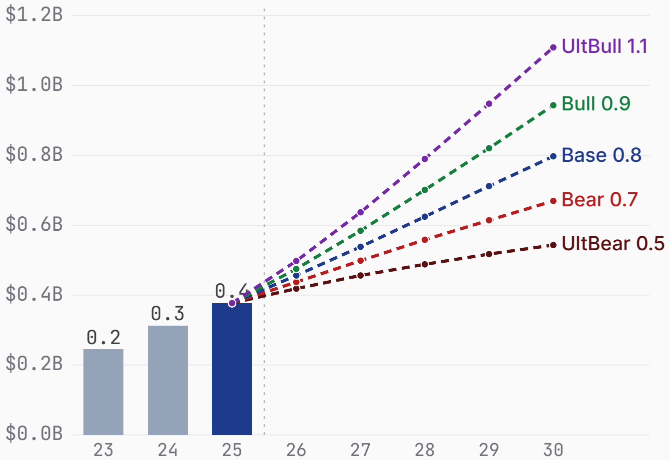
2 July 2026 · the snapshot

Twist Bioscience business snapshot

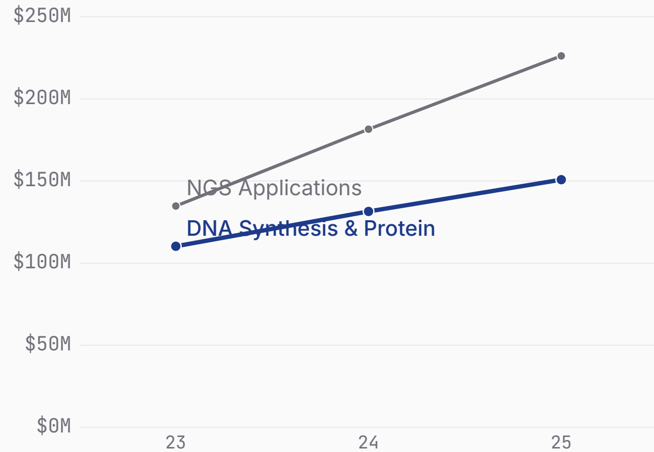
Bear $19.55 Base $37.12 Bull $73.69

FY23-FY25 history + FY26-FY30 scenario projections · fiscal years end Sept 30 · FY2025 + Q2 FY2026 results

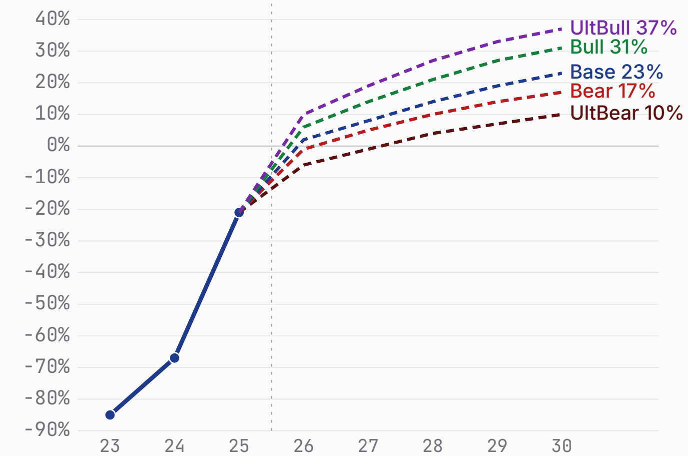
Revenue ($B) — history + scenarios to FY30

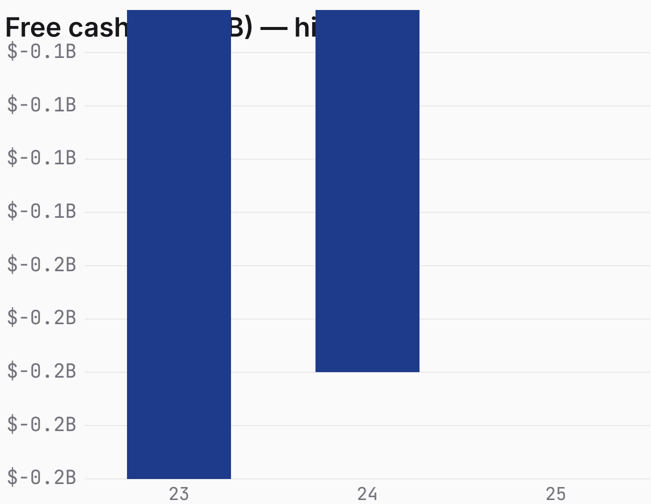


Revenue by segment ($M)

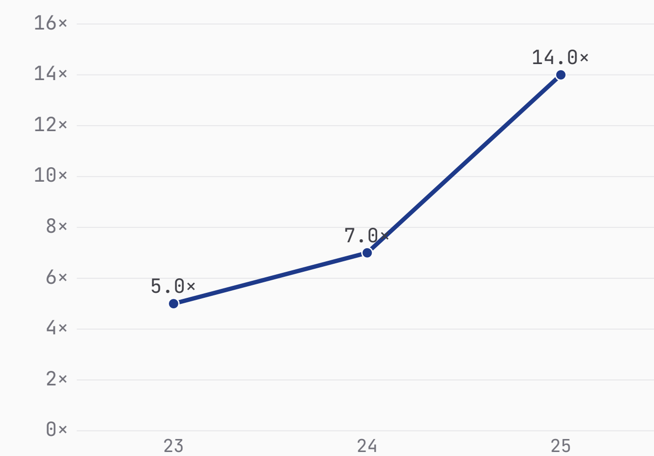


Op margin — history + scenarios

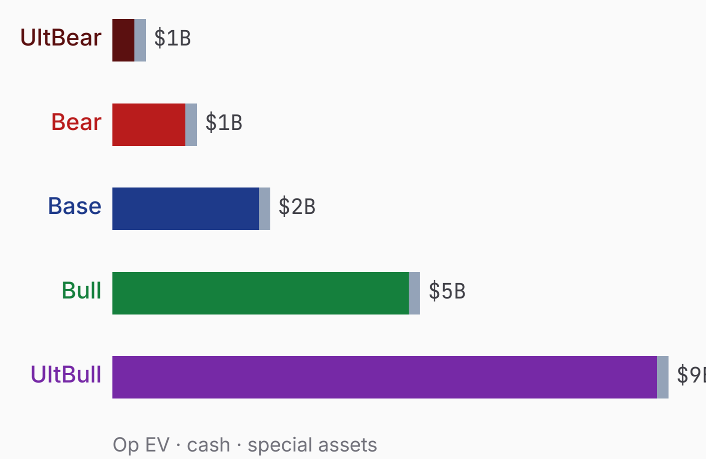




EV / Revenue — historical



Equity build (Op EV + net cash)



Sources: Twist Bioscience FY2025 (Sept 30) + Q2 FY2026 results (CIK 1581280), FY26 guidance. Revenue as a growth-rate path off FY25; FCF = revenue x an SBC-adjusted owner-FCF margin (starts negative — SBC keeps owner-FCF below the guided adj-EBITDA breakeven); Gordon terminal at scenario WACC. EV/sales vs DNA-synthesis / life-science-tools peers.

Twist Bioscience 7 Powers · the moat, scored

Bear $19.55 Base $37.12 Bull $73.69

Arena. High-throughput DNA synthesis (plus NGS tools) — Twist's silicon platform versus IDT/Danaher, GenScript, and enzymatic-synthesis challengers Ansa and DNA Script.

POWER AUDIT
dominant-power durability: medium

POWER AUDIT · 7 POWERS

Scale Economies — Silicon chip + the Wilsonville factory spread fixed cost across high volume — the core cost advantage, still ramping.

Network Economies — Essentially none — a DNA supplier's value does not rise with other customers' usage.

Counter-Positioning — Silicon synthesis is a structurally different, lower-cost method legacy column players cannot easily match without cannibalizing.

Switching Costs — Some workflow and validation lock-in for NGS and protein customers, but DNA is largely a fungible input across vendors.

Branding — Strong quality reputation for synthesis fidelity, but buyers are price- and spec-driven, not brand-driven.

Cornered Resource · thesis leans here — The proprietary silicon-synthesis process and IP — the throughput-per-cost engine rivals have not replicated at scale.

Process Power — Accumulated manufacturing know-how in miniaturized synthesis, but a process advantage rather than an unassailable one.

PEER SET

IDT (Danaher) · division of Danaher
Deep-pocketed synthesis leader inside Danaher; can fund price competition

GenScript 15% gr · public, well-funded
Broad gene-synthesis and CRO scale; mid-teens grower

Ansa Biotechnologies · venture-backed
Enzymatic synthesis — a potentially lower-cost long-DNA method

DNA Script · venture-backed
Enzymatic benchtop synthesis; alternative platform path

THREAT VECTORS · FALSIFIERS

scale_economies / pricing · IDT (Danaher), GenScript
falsifier: Gross margin failing to advance toward platform levels as well-capitalized rivals compete on price during the factory ramp.

cornered_resource erosion · Ansa, DNA Script
falsifier: Enzymatic synthesis reaching commercial cost/throughput parity, neutralizing the silicon platform's structural advantage.

optionality never lands · Atlas Data Storage
falsifier: DNA data storage failing to commercialize, leaving the spun-out equity stake worth little and removing the right-tail.

COMPETITIVE READ

Twist holds a real but medium-durability Power: a cornered silicon-synthesis process backed by scale economies the Wilsonville factory is still building. The audit verdict is a credible cost-leader, not yet an unassailable moat — DNA is partly fungible and a Danaher-owned IDT can fund a price war. The falsifier the bull must clear is the gross-margin ramp: if margins advance toward platform levels while 20%+ growth holds, the Power is durable; if pricing competition flattens the margin curve, it is a strong franchise priced as if it were a moat.

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2 July 2026 · the quantitative

Twist Bioscience show your work

Bear $19.55 Base $37.12 Bull $73.69 · Weighted $43.95 (-57.0%)

ULTRA BEAR $7.50

ASSUMPTIONS

5y revenue CAGR (%)	+7.6
Year-5 op margin (%)	10.0
WACC (%)	11.0
Terminal growth (%)	3.0
5y revenue CAGR (%)	+7.6
Starting revenue ($B)	0.38
Scenario probability (%)	13

5-YEAR DCF · $B

FY	Rev	Margin	FCF	PV FCF
FY26	0.42	-6%	-0.03	-0.03
FY27	0.46	-1%	-0.01	-0.01
FY28	0.49	+4%	+0.01	+0.01
FY29	0.52	+7%	+0.03	+0.02
FY30	0.54	+10%	+0.04	+0.03

Σ PV explicit FCF	+0.01
+ PV terminal (g 3.0%)	0.33

EQUITY BUILD · $B & PER SHARE

Operating EV	$0.34B
+ Cash & investments	$0.17B
= Equity value	$0.51B
FY30 shares (M)	68
= Expected per share	**$7.50**

FUTURE FAIR VALUE · IF SCENARIO PLAYS OUT

+5y	+10y	+15y	+20y
$12.64	$21.30	$35.88	$60.47
0.12×	0.21×	0.35×	0.59×

BEAR $19.55

ASSUMPTIONS

5y revenue CAGR (%)	+12.2
Year-5 op margin (%)	17.0
WACC (%)	10.0
Terminal growth (%)	3.5
5y revenue CAGR (%)	+12.2
Starting revenue ($B)	0.38
Scenario probability (%)	27

5-YEAR DCF · $B

FY	Rev	Margin	FCF	PV FCF
FY26	0.44	-1%	-0.01	-0.01
FY27	0.50	+5%	+0.01	+0.01
FY28	0.56	+10%	+0.04	+0.03
FY29	0.61	+14%	+0.07	+0.05
FY30	0.67	+17%	+0.10	+0.06

Σ PV explicit FCF	+0.15
+ PV terminal (g 3.5%)	0.99

EQUITY BUILD · $B & PER SHARE

Operating EV	$1.14B
+ Cash & investments	$0.17B
= Equity value	$1.31B
FY30 shares (M)	67
= Expected per share	**$19.55**

FUTURE FAIR VALUE · IF SCENARIO PLAYS OUT

+5y	+10y	+15y	+20y
$31.49	$50.71	$81.67	$131.52
0.31×	0.50×	0.80×	1.29×

BASE $37.12

ASSUMPTIONS

5y revenue CAGR (%)	+16.2
Year-5 op margin (%)	23.0
WACC (%)	9.5
Terminal growth (%)	4.0
5y revenue CAGR (%)	+16.2
Starting revenue ($B)	0.38
Scenario probability (%)	33

5-YEAR DCF · $B

FY	Rev	Margin	FCF	PV FCF
FY26	0.46	+2%	+0.00	+0.00
FY27	0.54	+8%	+0.03	+0.03
FY28	0.62	+14%	+0.07	+0.06
FY29	0.71	+19%	+0.12	+0.08
FY30	0.80	+23%	+0.17	+0.11

Σ PV explicit FCF	+0.27
+ PV terminal (g 4.0%)	2.01

EQUITY BUILD · $B & PER SHARE

Operating EV	$2.28B
+ Cash & investments	$0.17B
= Equity value	$2.45B
FY30 shares (M)	66
= Expected per share	**$37.12**

FUTURE FAIR VALUE · IF SCENARIO PLAYS OUT

+5y	+10y	+15y	+20y
$58.44	$91.99	$144.82	$227.98
0.57×	0.90×	1.42×	2.23×

BULL $73.69

ASSUMPTIONS

5y revenue CAGR (%)	+20.1
Year-5 op margin (%)	31.0
WACC (%)	9.0
Terminal growth (%)	4.5
5y revenue CAGR (%)	+20.1
Starting revenue ($B)	0.38
Scenario probability (%)	18

5-YEAR DCF · $B

FY	Rev	Margin	FCF	PV FCF
FY26	0.48	+6%	+0.02	+0.02
FY27	0.58	+14%	+0.07	+0.06
FY28	0.70	+21%	+0.13	+0.10
FY29	0.82	+27%	+0.20	+0.14
FY30	0.94	+31%	+0.27	+0.18

Σ PV explicit FCF	+0.50
+ PV terminal (g 4.5%)	4.12

EQUITY BUILD · $B & PER SHARE

Operating EV	$4.62B
+ Cash & investments	$0.17B
= Equity value	$4.79B
FY30 shares (M)	65
= Expected per share	**$73.69**

FUTURE FAIR VALUE · IF SCENARIO PLAYS OUT

+5y	+10y	+15y	+20y
$113.38	$174.45	$268.41	$412.99
1.11×	1.71×	2.63×	4.04×

ULTRA BULL $135.31

ASSUMPTIONS

5y revenue CAGR (%)	+24.1
Year-5 op margin (%)	37.0
WACC (%)	8.5
Terminal growth (%)	5.0
5y revenue CAGR (%)	+24.1
Starting revenue ($B)	0.38
Scenario probability (%)	9

5-YEAR DCF · $B

FY	Rev	Margin	FCF	PV FCF
FY26	0.50	+10%	+0.04	+0.04
FY27	0.64	+19%	+0.11	+0.09
FY28	0.79	+27%	+0.20	+0.15
FY29	0.95	+33%	+0.29	+0.21
FY30	1.11	+37%	+0.39	+0.26

Σ PV explicit FCF	+0.75
+ PV terminal (g 5.0%)	7.74

EQUITY BUILD · $B & PER SHARE

Operating EV	$8.49B
+ Cash & investments	$0.17B
= Equity value	$8.66B
FY30 shares (M)	64
= Expected per share	**$135.31**

FUTURE FAIR VALUE · IF SCENARIO PLAYS OUT

+5y	+10y	+15y	+20y
$203.46	$305.93	$460.02	$691.71
1.99×	2.99×	4.50×	6.76×

Twist Bioscience People · Opportunity · Context · Deal

Bear $19.55 Base $37.12 Bull $73.69

Underwrite the position the way a venture investor underwrites a round — across all four legs, public or private. **People** is the leg scored here (observable only); Opportunity, Context and Deal cross-reference the rest of the memo.

PEOPLE · WHO RUNS / OWNS / FUNDS IT

Emily Leproust · founder-led — 13 yrs in seat

2.4%
insider ownership

MEDIUM
key-person risk

Capital allocation (realized)
Co-founder Emily Leproust has been CEO since 2013 and Chair since 2018; co-founder Bill Banyai remains an executive and director (the DNA-data-storage "Atlas" spinout). The synthetic-DNA business is past early stage (~$377M revenue, +20%) and targeting adjusted-EBITDA breakeven by late FY2026, on ~$232M cash. Funding is via an active $200M ATM (ongoing dilution); no dividend or buyback.

Incentive alignment
Leproust's FY2025 compensation was ~$13.8M (equity-heavy; up on a large stock grant), with ~94% say-on-pay support. Insiders hold only ~2.4% of the company; alignment is through equity awards, not a founding control block. Insider trading is net selling under a 10b5-1 plan (much of it RSU tax withholding), no open-market buying.

GOVERNANCE FLAGS

- single class of common stock, one vote per share — no dual-class or super-voting; no controlling shareholder (largest holder ARK ~12%)
- combined Chair & CEO (Leproust), mitigated by a lead independent director (Robert Chess)
- classified / staggered board (three classes) — an entrenchment feature
- 7 of 9 directors independent; ongoing ATM dilution with no buyback to offset
- a securities class action (following a short-seller report) was allowed to proceed — an open litigation flag
- director interlock: a sitting director (Ragusa) is CEO of another company in the watchlist (GRAIL)

> PEOPLE READ
> A clean one-share-one-vote register with no controlling holder, a 7-of-9 independent board, and a lead independent director are the positives. Holding it at 3 are a combined Chair/CEO, a classified board, ongoing ATM dilution, and an open securities class action. Key-person risk is moderate — a founder Chair/CEO, but a second co-founder still active, an institutionalized board, and a business past early stage (~$377M revenue).

THE FOUR LEGS

People ●●●○○
observable composite · 1–5

Opportunity
The scenario distribution · the Page-3 business snapshot · the Arthur Indicator

Context
The 7 Powers analysis (Power Audit)

Deal
Open-market purchase = the deal: entry price vs. the scenario distribution (the -49.8% finding) + position sizing.

Twist Bioscience supporting analysis · disclaimers · glossary

Bear $19.55 Base $37.12 Bull $73.69

PUSHBACK · WHY THE BASE CASE IS TOO HARSH

1 **Cost-leader picks-and-shovels**
A cheaper way to write DNA — a 51%-GM platform selling shovels to every genomics buyer, not one drug.

2 **Real breakeven inflection**
Q4 FY2026 adj-EBITDA breakeven is real; 13 sequential growth quarters and the Wilsonville factory make the margin ramp credible.

3 **Free Atlas optionality**
Data storage spun out as Atlas in May 2025; Twist holds equity, so any commercialization is upside the DCF ignores.

4 **Multiple, not the business**
The bear case is about price, not execution — at ~14x sales even a well-run 16% grower is worth far less than spot.

5 **SBC gap to owner-FCF**
Adj-EBITDA flatters: heavy stock comp keeps owner-FCF below the headline, so 'breakeven' precedes positive owner cash flow.

FALSIFICATION TRIGGERS

Bull validation
Growth sustained at 20%+ with adj-EBITDA durably positive; gross margin marching toward platform levels; Atlas gaining traction.

Bear validation
Growth decelerating toward low-teens; the margin ramp stalling under IDT/GenScript pricing; SBC keeping owner-FCF negative well past the guided breakeven.

Reframe needed
Sustained price competition on gross margin, or a biotech-funding winter cutting orders, moves this from rich-but-executing to challenged.

DISCLAIMERS · PLEASE READ BEFORE USING THIS DOCUMENT

GLOSSARY · CONCEPTS REFERENCED IN THE NARRATIVE

Silicon DNA synthesis — Twist's core platform — DNA written on a miniaturized silicon chip, enabling high throughput at low cost per base versus older column methods.

SynBio / NGS segments — The two demand areas: synthetic biology (DNA Synthesis and Protein Solutions) and next-generation sequencing applications (target enrichment, library prep).

Factory of the Future — The new Wilsonville, Oregon plant built to scale capacity and step up gross margin — the engine behind the modeled margin ramp.

Atlas (DNA data storage) — The DNA-data-storage venture spun out as independent Atlas Data Storage in May 2025; Twist retains equity, so it is now an external call option, not in-house.

Adj-EBITDA vs owner-FCF — Adj-EBITDA adds back stock comp (breakeven Q4 FY2026); owner-FCF subtracts real SBC + capex, so it turns positive later.

SBC — Stock-based compensation — a real cost to owners via dilution that adj-EBITDA excludes; deducted here from owner FCF.